GRADE 1 · INTRO DEBUGGING

The Extra Block

Take out the block that goes too far

Name: .... Date: ....

★ **I can fix a program by removing an extra block.**

Too far!

Sometimes a program has one EXTRA block, so Bit goes too far and overshoots the target. Run the code, find the extra block, and cross it out to fix the program.

Bit's number path

0	1	2	3	4	5

Program 1 — target is 4

start at 1

forward 2

forward 1

forward 1

1 Run it — too far

Run Program 1 from 1. What number does Bit land on? It goes past the target 4.

2 Remove the extra block

Cross out ONE block so Bit lands exactly on 4. Which block did you remove? Run it again — what number now?

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Program 2 — target is 1

start at 0

forward 2

back 1

back 1

3 You try — remove to fix

Run Program 2 from 0. It lands on 0, not 1. Cross out ONE block so Bit lands on 1. Which block did you remove, and where does Bit land?

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Big idea

If a program goes too far, it may have an extra block. Take out the one extra step so Bit lands right on the target.

TEACHER GUIDE · GRADE 1 · INTRO DEBUGGING

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Quick facilitation notes & answer key

What this worksheet teaches

Students fix a bug where the program goes PAST the target because it has one extra block; they cross out the extra step and run again. Removing a step to fix the result is a kind of debugging. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: tape a number line 0 to 5 on the floor and mark the target, so students can walk the buggy program and feel Bit miss.

How to lead it (about 20 minutes)

1. Show them (you do). Run a program that overshoots the target — it has one step too many.
2. Do one together. Exercise 1 — Program 1 lands on 5, past 4.
3. Let them try. Students cross out one block to land exactly on the target (Ex 2), then fix Program 2 (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Program 1: 1 → 3 → 4 → 5. Bit lands on 5, past the target 4.

Exercise 2 — remove one "forward 1": 1 → 3 → 4. Bit lands on 4.

Exercise 3 — Program 2: 0, forward 2 → 2, back 1 → 1, back 1 → 0. Remove one "back 1": 0 → 2 → 1. Bit lands on 1.

If students get stuck — and ways to adjust

Watch for: changing a block instead of removing one. Here the fix is to take a step away.

Make it easier: walk the program and notice the step that goes one too far.

Make it harder: ask how many steps past the target Bit went, and why that points to one extra block.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a program, change a block to fix it, and explain what the change did.

You'll know it worked when

The child removes the one extra block so Bit lands exactly on the target.